Lesson 2026/09/30

livio

2026-09-30

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Import

```
dati <- read.csv("data/students.csv")

str(dati)
```

```
## 'data.frame':    443 obs. of  4 variables:
##  $ X                   : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ student_id          : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ classes_attended    : int  7 5 7 5 4 9 6 5 6 7 ...
##  $ assignments_submitted: int  6 2 6 7 6 9 9 2 7 8 ...
```

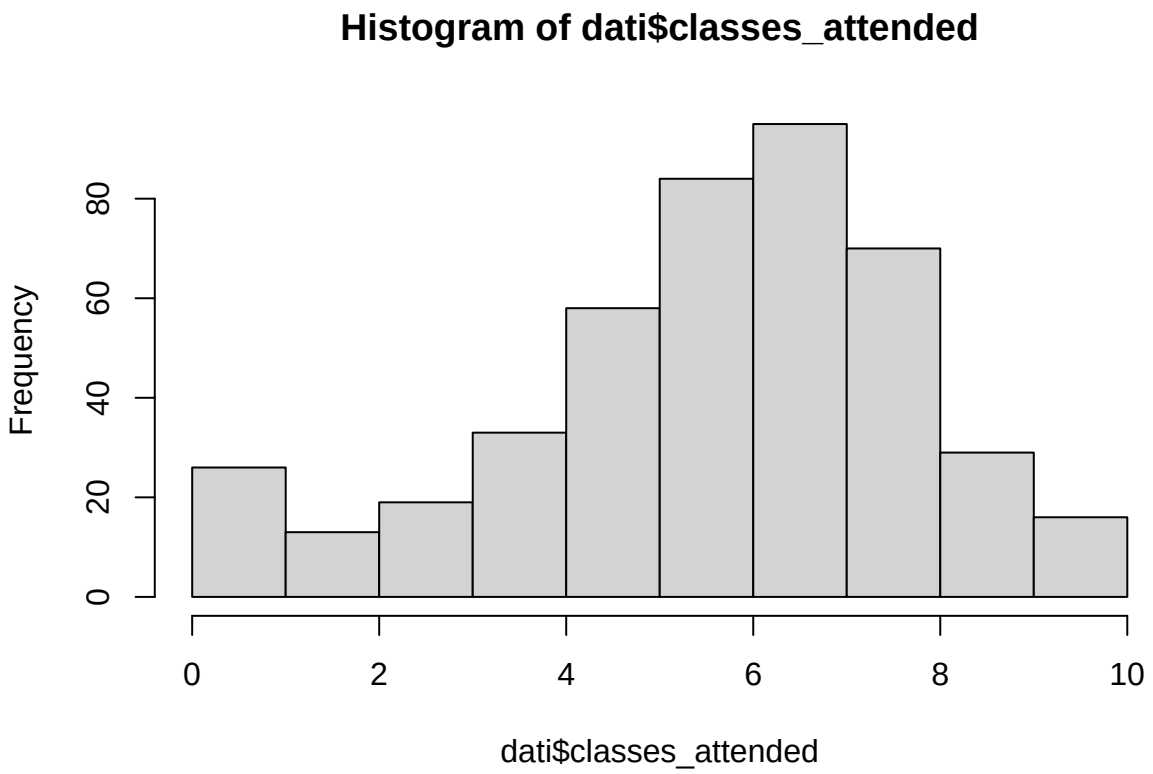
summary

```
summary(dati)
```

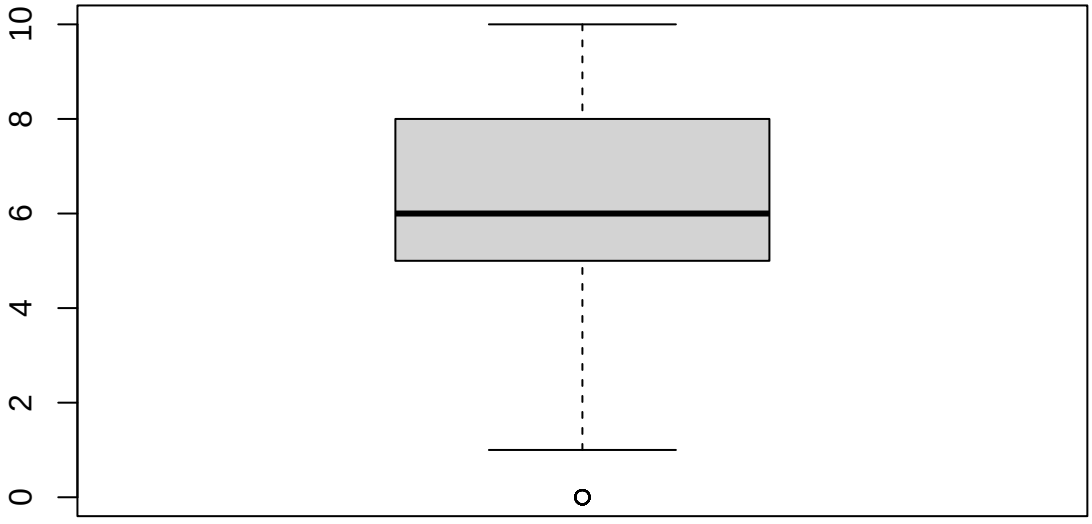
```
##        X           student_id    classes_attended assignments_submitted
##  Min.   :  1.0   Min.   :  1.0   Min.   : 0.000   Min.   : 0.000
##  1st Qu.:111.5   1st Qu.:111.5   1st Qu.: 5.000   1st Qu.: 5.000
##  Median :222.0   Median :222.0   Median : 6.000   Median : 7.000
##  Mean   :222.0   Mean   :222.0   Mean   : 6.023   Mean   : 6.765
##  3rd Qu.:332.5   3rd Qu.:332.5   3rd Qu.: 8.000   3rd Qu.: 9.000
##  Max.   :443.0   Max.   :443.0   Max.   :10.000   Max.   :12.000
```

plot

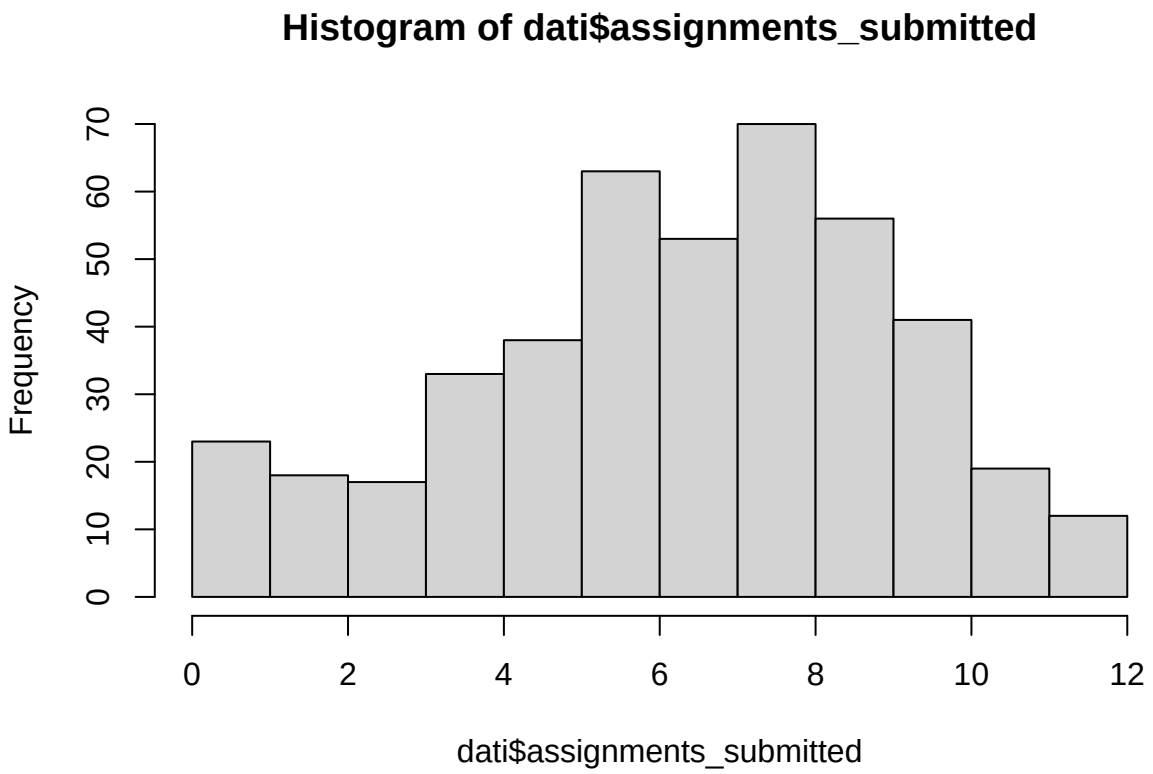
```
hist(dati$classes_attended)
```



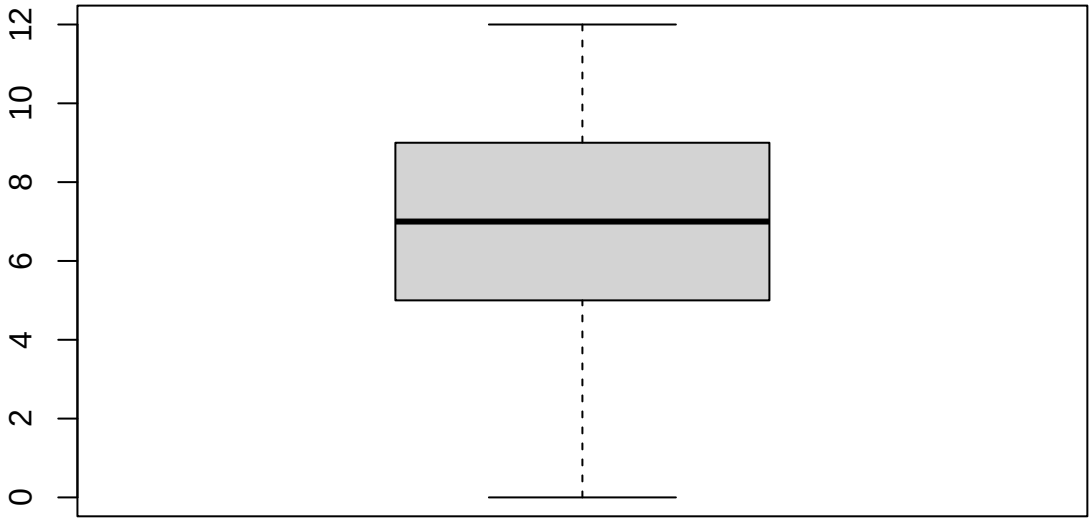

```
boxplot(dati$classes_attended)
```




```
hist(dati$assignments_submitted)
```




```
boxplot(dati$assignments_submitted)
```

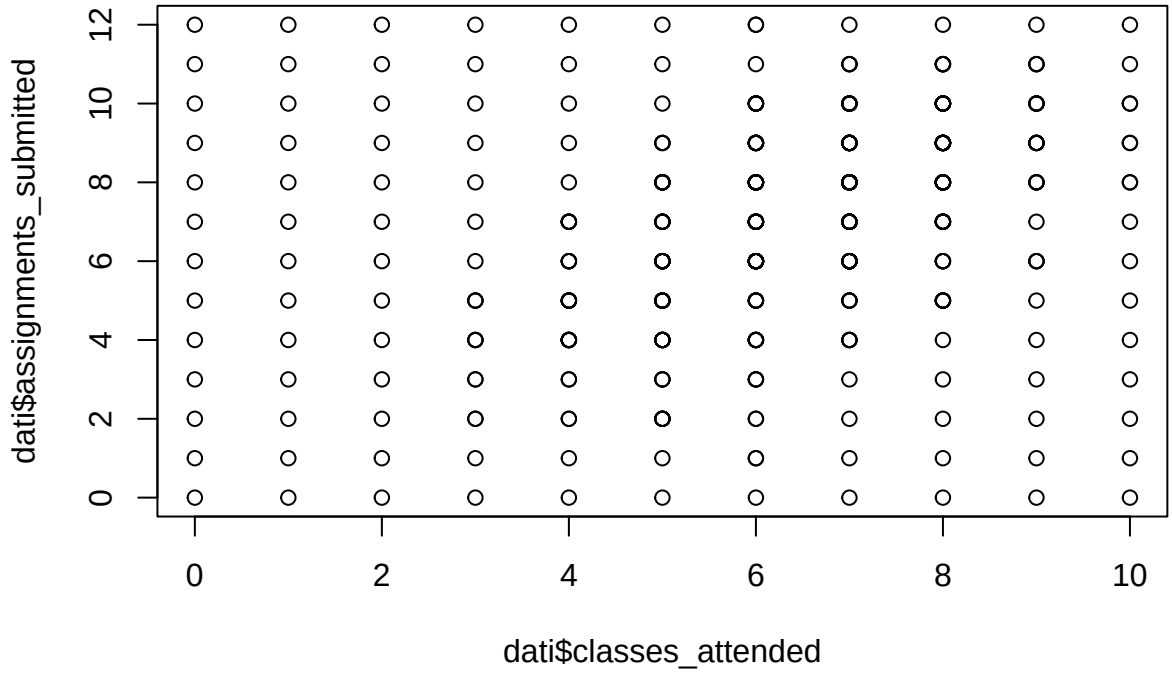



```
table(dati$classes_attended,dati$assignments_submitted)
```

```
##
##      0  1  2  3  4  5  6  7  8  9 10 11 12
##   0  1  1  1  1  1  1  1  1  1  1  1  1  1
##   1  1  1  1  1  1  1  1  1  1  1  1  1  1
##   2  1  1  1  1  1  1  1  1  1  1  1  1  1
##   3  1  1  2  2  3  3  1  1  1  1  1  1  1
##   4  1  1  2  2  7  6  4  5  1  1  1  1  1
##   5  1  1  5  3  7  9 12  6  8  3  1  1  1
##   6  1  2  2  3  6  5 18 13 14 10  7  1  2
```

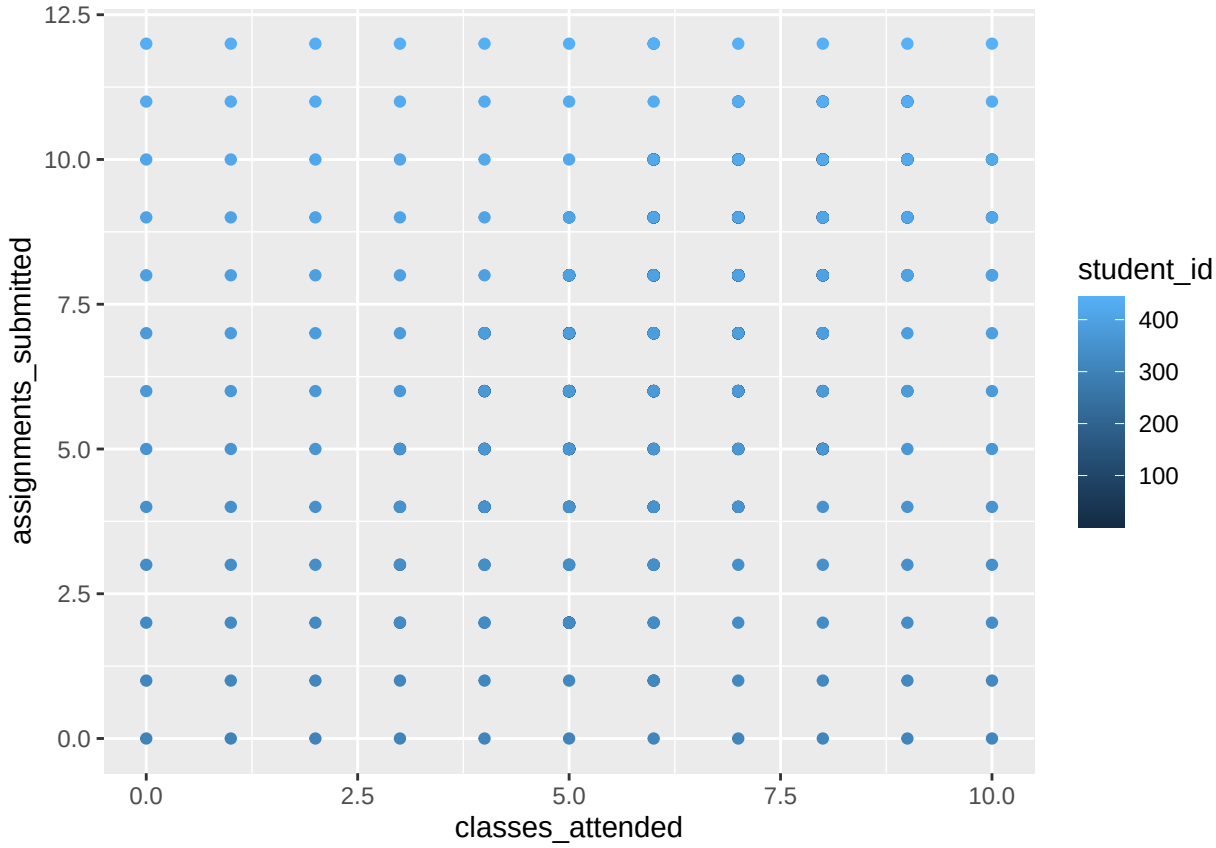
```
##    7   1  1  1  1  4  4 17 15 24 14  9  3  1
##    8   1  1  1  1  1  6  4  8 13 16 12  5  1
##    9   1  1  1  1  1  1  3  1  4  6  5  3  1
##    10  1  1  1  1  1  1  1  1  2  2  2  1  1
```

```
plot(dati$classes_attended,dati$assignments_submitted)
```

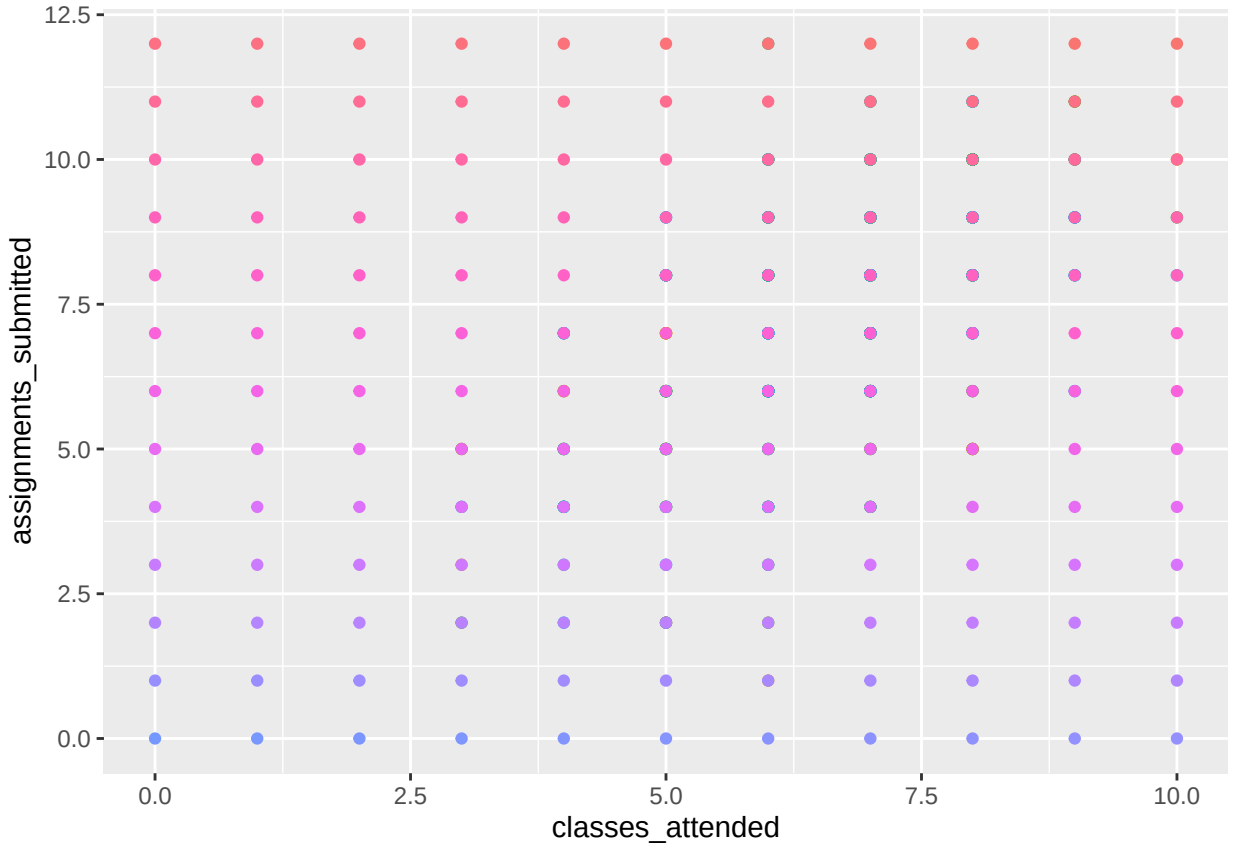



```
library(ggplot2)

ggplot(dati,aes(x=classes_attended,y=assignments_submitted,colour = student_id))+geom_point()
```

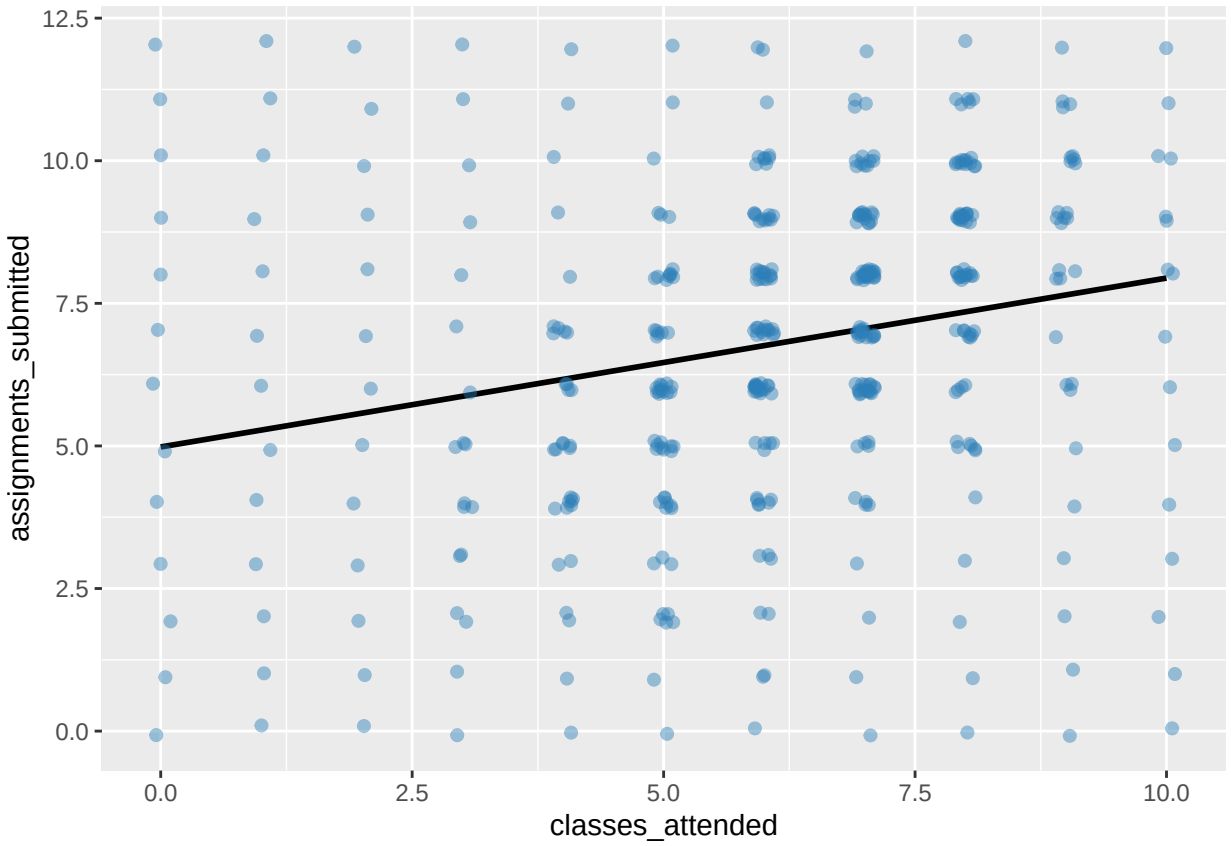



```
dati$student_id=factor(dati$student_id)
ggplot(dati,aes(x=classes_attended,y=assignments_submitted,colour = student_id))+geom_point()+theme(leg
```




```
ggplot(dati,aes(x=classes_attended,y=assignments_submitted))+
 geom_smooth(aes(group = 1), method = "lm", se = FALSE,
             colour = "black") +geom_jitter(
    width = 0.10,
    height = 0.10,
    size = 1.8,
    alpha = 0.45,
    colour = "#2C7FB8"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


12.5
10.0
7.5
5.0
2.5
0.0
assignments_submitted
0.0
2.5
5.0
7.5
10.0
classes_attended